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June 27, 2026

To: The Guest Editors,

IEEE Transactions on Network Science and Engineering,

Special Issue on *Tutorials and Surveys on AI and Quantum-enabled Learning for Future Wireless Networks*

(Prof. Trung Q. Duong, Prof. Ruidong Li, Dr. Bhaskara Narottama, Prof. George K. Karagiannis)

Re: Submission of the manuscript “*Quantum and Quantum-Inspired Graph Neural Networks for Dynamic Wireless Network Topologies: A Tutorial and Survey with a Reproducible LEO Case Study*”

Dear Guest Editors,

I am pleased to submit the above manuscript for consideration in the Special Issue on *Tutorials and Surveys on AI and Quantum-enabled Learning for Future Wireless Networks*. The paper is prepared as a combined **tutorial and survey**, the contribution type the call solicits, and it addresses several of the listed topics directly, including *quantum graph neural networks*, *data-driven quantum algorithms for dynamic topologies*, *near-term quantum computing for routing in non-terrestrial networks*, and *non-centralized learning of classical and quantum data*.

What the paper offers. The manuscript is organized around a single unifying idea: classical graph neural networks, graph diffusion, and continuous-time quantum walks are the same object, a propagation operator on the graph Laplacian, so the only thing that changes between them is the spreading physics (diffusive for the heat kernel, ballistic for the quantum walk). The method is therefore explicit and simple to state: write every model as a propagation operator acting on graph signals, vary only that operator on a parameter-matched backbone, and measure honestly. From this view the paper (i) surveys the classical, quantum, quantum-inspired, and dynamic-graph branches and their use in wireless and non-terrestrial networks, organized by operator rather than by author; (ii) teaches, with pseudocode, a formal result (a proposition deriving the diffusive-versus-ballistic spreading laws exactly on the line graph), and a worked example, how to build a parameter-matched quantum-inspired GNN layer that runs on classical hardware today; and (iii) demonstrates the framework on a fully reproducible low Earth orbit (LEO) case study.

The central finding is honest and, I believe, timely for this Special Issue. On a node-potential task over a dynamic LEO graph, global propagation beats local message passing by a wide margin at every scale, and the quantum-inspired and classical operators trade places as the constellation grows: the quantum-inspired walk loses at 66 satellites, ties at 264, and becomes the best operator at 1584 satellites, exactly the diameter-driven trend its ballistic spreading predicts. The paper reports this conditional, scale-dependent quantum advantage with its caveats rather than as a blanket claim, and adds a NISQ-noise study (PennyLane) showing how device noise erodes the mechanism. A dedicated section on evaluation pitfalls, scale-coupled normalization and single-seed reporting, distills a reusable checklist that should be useful to the community independently of the case study.

Reproducibility. Every figure and number in the paper is regenerated from seeded runs by the companion code at github.com/haodpsut/qi-gnn-dynamic-wireless. All experiments are parameter-matched and multi-seed.

Originality. This manuscript is original, has not been published before, and is not under consideration elsewhere. The LEO topology generator reuses my open-source simulator, while the node-potential task, the experiments, and all results here are new.

I confirm there are no conflicts of interest with the Guest Editors that would prevent a fair review, and I am happy to suggest or exclude reviewers if helpful.

Thank you for your time and consideration.

Sincerely,

Phuc Hao Do

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